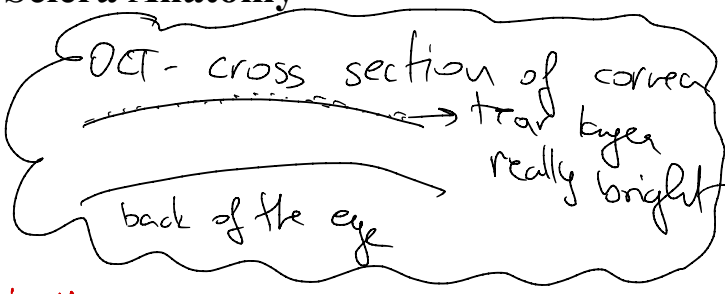
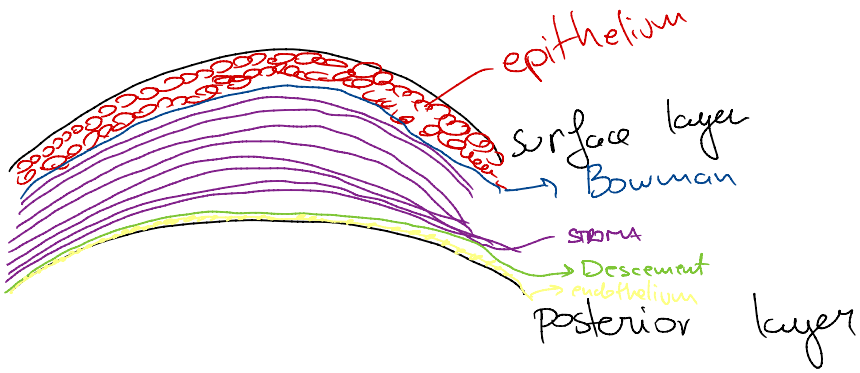
Cornea, Conjunctiva, and Sclera Anatomy

Images used: 2.2, 2.5, 2.10, 2.1, 2.8, 2.9, 2.12, 2.14



I. Corneal layers 5 layers of cornea →



A. ~~Epithelium~~

1. 5-7 cells thick
2. Surface epithelial cells (squamous cell layer)
    a. Secretes a glycocalyx that adjoins the mucin layer of the tear film.
    b. Has projections on the outer surface that enhance tear stability
3. Tight junctions (zonula occludens) join the lateral cell walls
    keep the water out
4. Clinical correlate: fluorescein dye
    should not penetrate in the cornea but when abnormalities then it stains

5. The basal layer of the epithelium connects attaches onto the underlying tissue by hemidesmosomes.
   commonly after abrasion, epithelium will grow back. But hemidesmoses, do NOT grow back correctly.
6. Clinical correlate: recurrent corneal erosion

B. Bowman layer

1. Dense sheet of interwoven collagen fibrils
   hard to see, because it is so thin. No cells, collagen.

2. Acellular

3. Does not regenerate
   Doesn't regenerate. Nevers comes back. Fills with epithelium or stroma → scar.

C. Stroma

1. About 90% of the corneal thickness

2. Composed of collagen fibrils, keratocytes and extracellular ground substance
   ↳ if not then it's gonna be white scar. Fibers not regular

3. The very regular arrangement and diameter of the stromal fibrils, as well as the restriction of the distance between fibrils, contribute to stromal transparency.

4. Clinical correlate: corneal opacification

D. Descemet membrane

1. The basement membrane of the corneal endothelium

2. It is produced throughout life and therefore thickens over a lifetime

3. Resistant to damage, but can be regenerated if damaged

4. As Descemet membrane ends at the limbus, it is called Schwalbe line.

E. Endothelium - pump water out of cornea? into

1. Single layer of flattened hexagonal cells

2. Ionic pumps help maintain hydration of the cornea

3. Endothelial cells to not divide and replicate - They spread out and gets bigger to fill empty space.

highly sensitive are of the most in the body.

II. Corneal innervation

A. Branches of the long and short ciliary nerves.

B. Plays a role in corneal structure and function

If you have

Diabetes does damage to the nerve. Will get staining in the eye because of the damage.

III. Corneal nutrition

A. The cornea gets oxygen primary from atmospheric oxygen that is dissolved in the tears.

1. Small amounts from the aqueous and limbal capillaries.

B. It obtains most nutrients from the aqueous.

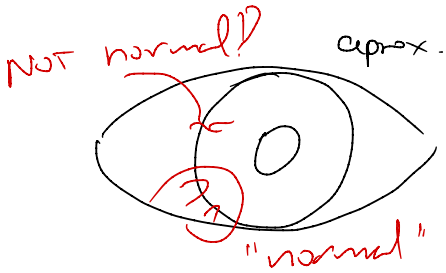
1. It also gets some from the conjunctival and episcleral vessels in the limbus.

C. Clinical correlate: corneal neovascularization → Posterior embryotoxin, arcus

↳ new blood vessels

we do NOT want to have vessels in the cornea at all.

By CL (contact lenses) they can start growing in because they need oxygen.



IV. Scleral anatomy – avascular

A. Functions

1. Maintains the shape of the globe
2. Resistance to external forces
3. Attachment for the extraocular muscles

B. Thick, dense connective tissue

irregular give the white color/opeque.



C. The sclera is continuous with the corneal stroma at the limbus

D. Scleral spur

1. Circularly oriented collagen fibrils at the inner, anterior aspect of the sclera.
2. Origin of the ciliary body muscle fibers.

E. Lamina cribrosa

↳ see it in optic nerve → bright ◯

1. Scleral tissue that bridges the foramen in with the optic nerve passes.

F. Blood supply

episclera in choroidal blood vessels.

V. Episclera

A. Loose very vascularized connective tissue just outer to the sclera

B. Blood supply to sclera and peripheral cornea

VI. Tenon capsule

so thin; between episclera and conjuctiva

A. This fibrous sheet of connective tissue outer to the episclera

B. Protects the globe - main purpose!

VII. Conjunctiva

A. Bulbar conjunctiva – sclera

B. Fornix

C. Palpebral conjunctiva – lids

VIII. Limbus - junction between CORNEA & SCLERA

A. Corneoscleral junction

B. Boundary: where Bowman layer and Descemet membrane end